$f \in \mathfrak{N}(W)$ *satisfy*

$$A(f) \leqslant CL(f)$$

for $C = N_0 \delta^2$. (See 6.8 for the proof.)

2.3.G **Remark:** Given a finite presentation $\{G \mid W\}$ there is an obvious algorithm for writing down all $f \in \mathfrak{N}(W)$ with $A_1 \leqslant A(f) \leqslant A_2$ and $L_1 \leqslant L(f) \leqslant L_2$ for given integers A_1, A_2, L_1 and L_2. Thus one obtains an effective (but not very practical) criterion for the word hyperbolicity of finitely presented groups Γ.

2.4 **Convex manifolds:** Consider a Riemannian manifold X with a convex (possibly empty) boundary. For example, X may be a *convex* subset in a complete manifold $X' \supset X$ without boundary, where a subset X in a Riemannian manifold is called convex if the intersection of X with every geodesic segment is connected. Recall that the distance between two points x and y in X is defined as the infimum of lengths of curves in X between x and y.

Let X be simply connected and complete as a metric space and recall the following well known (and easy to prove) geometric fact.

2.4.A **Theorem:** *The sectional curvature* $K(X)$ *satisfies* $K(X) \leqslant 0$ *if and only if every smooth closed curve* F *in* X *bounds a disk* D *in* X *satisfying*

$$\text{area } D \leqslant (4\pi)^{-1}(\text{length } F)^2 \tag{4}$$

Furthermore, if the curvature $K(X)$ *is strictly negative,* $K(X) \leqslant -\epsilon^2 < 0$, *then every curve* F *bounds a disk* D, *such that*

$$\text{area } D \leqslant \epsilon^{-1} \text{ length } F. \tag{5}$$

Remark: Notice that the inequality (4) is sharp as round disks D in $\mathbb{R}^2$ have area $D = (4\pi)^{-1}(\text{length } \partial D)^2$.

2.4.B <u>Definition</u>: Call X *convex* (see [Gr1] for a more general definition) if it is complete simply connected with a convex boundary and $K(X) \leqslant 0$.

It is well known (and easy to prove) that the strict inequality $K(X) \leqslant -\epsilon^2 < 0$ implies the hyperbolicity of X. In fact, a convex space X is (easily seen to be) hyperbolic if every smooth closed curve F in X of length $\ell \geqslant \ell_0$ for some constant $\ell_0 = \ell_0(X) > 0$ satisfies either of the following two conditions.

(1) F bounds a disk D, such that area $D \leqslant C_0$ length F for some $C_0 = C_0(X) > 0$.

(2) F bounds a disk D, such that

$$\text{area } D \leqslant [(4\pi)^{-1} - \epsilon_0](\text{length } F)^2$$

for some $\epsilon_0 = \epsilon_0(X) > 0$.

2.4.C <u>Remark</u>: The δ-inequality (see 1.1) for $K < -\epsilon^2 < 0$ is a trivial consequence of the CAT-*inequality* (comparison inequality of Alexandrov-Toponogov). This inequality applies to a pointed metric space (X, x_0) and triples of points x, y and z in X where z lies *between* x and y. That is

$$|z-x| + |z-y| = |x-y|.$$

CAT relates distances between the points in X to the distances in the *model* space (X', x_0'), which is a complete simply connected surface with a *constant* curvature χ for a given $\chi \in \mathbb{R}$. In other words, this X' is a sphere S^2, the plane $\mathbb{R}^2$ or the hyperbolic plane H^2. Points x', y' and z' in X' are called *comparison* points (for x, y, z) if

$$|x'| = |x|, \ |y'| = |y|, \ |x'-y'| = |x-y|$$

and

$$|x'-z'| = |x-z|, \ |y'-z'| = |y-z|.$$

The CAT($\varkappa$)-inequality claims, by definition, that the comparison points exist (for given x, y and z) and $|z'| \geqslant |z|$.

One of the basic results in Riemannian geometry is the following

2.4.D **<u>Theorem</u> (Riemann-Cartan-Alexandrov-Toponogov):** *A Riemannian manifold X with convex boundary has* $K(X) \leqslant \varkappa$ *if and only if each point* $x_0 \in X$ *admits a neighborhood* $U_0 \subset X$ *of* x_0 *where every three points satisfy* CAT($\varkappa$). *Furthermore, every three points in a convex manifold with* $K(X) \leqslant \varkappa \leqslant 0$ *satisfy* CAT($\varkappa$).

2.4.E **<u>Remark</u>:** The CAT($\varkappa$)-inequality for $\varkappa < 0$ easily implies the following sharpening of the δ-inequality.

Let X *be a geodesic space satisfying* CAT($\varkappa$) *for some* $\varkappa < 0$. (That is every three points satisfy CAT for every reference point in X.) *Take three points* x, y *and* z, *such that*

$$(x.y) \leqslant (x.z) \leqslant (y.z),$$

and let $d = (x.y) - (y.z)$. *Then*

$$(**) \qquad (x.y) \geqslant (x.z) - \delta$$

for some function $\delta = \delta(\varkappa, d) \geqslant 0$ *satisfying* $\delta \leqslant d \exp((-d/4)\sqrt{|\varkappa|})$.

2.5 **<u>Isometry group</u>:** Let Γ be a subgroup of the isometry group Is X. Then for every bounded open subset $U \subset X$ there obviously exists a unique left invariant metric on Γ, such that $|\gamma| = \sup_{x \in U} |x - \gamma(x)|_X$ for all $\gamma \in \Gamma$. If X is hyperbolic, then an obvious consideration shows this metric on Γ also is hyperbolic. In such a case the *orbit map* $\Gamma \longrightarrow X$ defined by $x_0 \longmapsto \gamma x_0$ for a given point $x_0 \in X$ continuously extends to an embedding between the hyperbolic boundaries,

$$\partial\Gamma \longrightarrow \partial X,$$

and the image $\partial\Gamma \subset \partial X$ is called the *limit set of* Γ.

If X is convex then the hyperbolic boundary ∂X of X is naturally homeomorphic to the space of *geodesic rays* in X issuing from a fixed point $x_0 \in X$, where a ray in X by definition is an isometric embedding $\mathbb{R}_+ \longrightarrow X$. This description of ∂X provides an embedding of ∂X into the tangent unit sphere $S^{n-1}_{x_0}$, $n = \dim X$, where each ray r is mapped to the tangent vector to r at x_0. The map $\partial X \longrightarrow S^{n-1}_{x_0}$ is *onto* if and only if the ordinary boundary of X is empty as an obvious consideration shows.

2.6 **Cocompact groups:** A group Γ acting on X is called *cocompact* if the quotient space X/Γ is compact. Every discrete cocompact isometry group obviously is finitely presented and (by an easy argument) *Γ is word hyperbolic if and only if X is hyperbolic.* For example, the fundamental group Γ of a *compact* Riemannian manifold V is word hyperbolic if and only if the universal covering X of V is hyperbolic for the *induced Riemannian metric* in X. Observe that this metric is, in fact, the maximal metric on X for which the covering map $X \longrightarrow V$ is locally isometric.

Let us indicate two unsolved problems.

(A) Let Γ be word hyperbolic. Does there exist a convex manifold X which admits a discrete cocompact isometry group isomorphic to Γ?

(B) Let Γ be a discrete cocompact isometry group of a convex manifold X, such that every Abelian subgroup in Γ is virtually cyclic (i.e. a finite extension of a cyclic group). Is Γ hyperbolic? (One can easily show that if Γ is not hyperbolic then there exists an isometric map $\mathbb{R}^2 \longrightarrow X$. But the question is if there is an isometric map $\mathbb{R}^2 \longrightarrow X$ for which the image of $\mathbb{R}^2$ in X/Γ is compact, compare 4.7.)

2.7 **Symmetric spaces:** Let X be a symmetric space of non-compact type without Euclidean factors (i.e. X is not isometric to $X_0 \times \mathbb{R}^k$ for $k > 0$). Then the isometry group Is X is a semisimple Lie group without compact factors. One knows that Is X contains a torsion free cocompact discrete subgroup Γ and every such Γ is residually finite. If rank X = 1 (i.e. there is no isometric map $\mathbb{R}^2 \longrightarrow X$) then Γ is word hyperbolic. Thus, by taking $X = O(n,1)/O(n)$, one obtains for every $n = 2,3,\ldots$, a compact aspherical manifold V, namely $V = X/\Gamma$, whose fundamental group Γ is hyperbolic and residually finite. Notice that the symmetric spaces $U(n,1)/U(n)$ and $Sp(n,1)/Sp(n)$ have dimensions 2n and 4n respectively. The group $Sp(n,1)$ has an additional remarkable feature, namely *Kazdan's T-property* (see 5.6) which is inherited by all lattices Γ in $Sp(n,1)$ and has a profound effect on algebraic properties of Γ. Notice that no *known* purely algebraic construction delivers infinite groups Γ with the T-property.

If X has rank $\geqslant 2$ then no cocompact Γ is hyperbolic. However, there are many word hyperbolic groups Γ operating on X, such as hyperbolic *reflection groups* Γ (see 4.6) which isometrically act on $X = O(p,q)/O(p) \times O(q)$. One usually proves the hyperbolicity of a group Γ operating on X by constructing a hyperbolic (e.g. convex) Γ-invariant submanifold $X_0 \subset X$, such that X_0/Γ is compact.

§3. The action of Γ on $\partial\Gamma$

Consider a hyperbolic (metric) group Γ and look at the hyperbolic boundary $\partial\Gamma$ of Γ. The simplest example is that of a free group G_k on k generators which isometrically acts on the regular infinite tree X with 2k edges of unit length at every vertex. It is obvious that $\partial F_k = \partial X$ and that ∂X is (homeomorphic to) a Cantor set provided $k \geqslant 2$. If $k = 1$, then the boundary consists of two isolated points.

3.1 **Elementary and non-elementary groups**: Let us classify hyperbolic groups according to the cardinality of the boundary.

Case 1: The boundary $\partial\Gamma$ is empty. In this case Γ is bounded as a metric space (see 8.1, 8.2).

Case 2: The boundary $\partial\Gamma$ consists of a single point. Such groups Γ are called *parabolic*. Observe that every infinite group Γ admits a parabolic metric. Namely, start with any unbounded metric $|\ |$ and then take the (hyperbolic!) metric $\log(1 + |\ |)$ (see 1.2). It is easy to see that $(\Gamma, \log(1 + |\ |))$ is parabolic.

Case 3: The boundary of Γ consists of two points. Then (see 8.1) Γ contains an infinite cyclic subgroup, say $\Gamma_0 \subset \Gamma$ which is *cobounded* in Γ. In particular, if Γ is word hyperbolic then Γ is a finite extension of Γ_0.

Recall that a subset $X_0 \subset X$ is cobounded if there is constant $d = d(X_0)$ such that $\mathrm{dist}(x, X_0) \leqslant d$ for all $x \in X$.

The groups in these three cases are called *elementary* hyperbolic groups.

Case 4: The boundary $\partial\Gamma$ contains at least three points. Then (see 8.2) the boundary is infinite uncountable and Γ contains an isomorphic copy of the free semigroup F_2^+ with two generators. In fact, there are two elements α_0 and α_1 in Γ, such that the map of the set of finite diadic sequences into Γ given by

$0 1 1 0 1 \ldots \longmapsto \alpha_0\alpha_1\alpha_1\alpha_0\alpha_1\ldots$ is injective and extends to an injective continuous map of the Cantor set $\{0,1\}^\infty$ to ∂X. (See 8.2.)

__Case 4'__: Call Γ *quasiparabolic* if there exists a point $p \in \partial X$ which is fixed under Γ. If Γ is *not* quasiparabolic as well as non-elementary, then the above α_0 and α_1 can be chosen freely independent in Γ and such that the monomorphism of the free group $F_2 = F(\alpha_0, \alpha_1)$ into Γ extends to a topological embedding $F_2 \cup \partial F_2 \longrightarrow \Gamma \cup \partial\Gamma$, where F_2 is given the usual word metric. (See 8.2.)

3.1.A __Remarks__:

(a) If Γ is *word hyperbolic* then the boundary $\partial\Gamma$ is a *compact finite dimensional* space (see 7.6.) and isomorphisms $\Gamma_1 \longrightarrow \Gamma_2$ between such groups continuously extend to homeomorphisms between their boundaries (see 7.2.H.). Thus every topological invariant of $\partial\Gamma$ is, in fact, an invariant of Γ. On the other hand a non-bijective homomorphism α: $\Gamma_1 \longrightarrow \Gamma_2$ does not necessarily extend to the boundary even if we assume α injective. However, there always exists a natural discontinuous map, called the *Furstenberg map*, sending a "large portion" of $\partial\Gamma_1$ to $\partial\Gamma_2$ (see [Fu]).

(b) If Γ is a subgroup in a word hyperbolic group, then it can not be parabolic or quasiparabolic. (See 8.1.) It follows that every such Γ contains F_2 unless Γ is a finite extension of a cyclic group.

3.2 __Group__ Γ __with__ dim $\partial\Gamma = 0$: Take two metric groups Γ_1 and Γ_2, let $\Gamma_i' \subset \Gamma_i$, $i = 1,2$, be subgroups and let $\Gamma_1' \longleftrightarrow \Gamma_2'$ be an isometric isomorphism. Then we consider the amalgamated product $\Gamma = \Gamma_1 \underset{\Gamma'}{*} \Gamma_2$ for $\Gamma' = \Gamma_1' = \Gamma_2'$ with the maximal metric in Γ for which the inclusion homomorphisms $\Gamma_i \longrightarrow \Gamma$, $i = 1,2$, are *short*, where a map f: $X \longrightarrow Y$ is called *short* if $|f(x_1)-f(x_2)| \leq |x_1-x_2|$ for all x_1 and x_2 in X. If the group Γ_1 and Γ_2 are hyperbolic and Γ' is bounded, then it is easy to see that Γ is hyperbolic and the topological dimension of $\partial\Gamma$

satisfies

$$\dim \partial\Gamma = \max(\dim \partial\Gamma_1, \dim \partial\Gamma_2).$$

Furthermore, the boundary $\partial\Gamma$ is infinite (and hence $\partial\Gamma$ is non-elementary), provided $\#(\Gamma_1/\Gamma_1') \geq 2$ and $\#(\Gamma_2/\Gamma_2') \geq 3$. For example, if the groups Γ_1 and Γ_2 are elementary (e.g. bounded) then Γ has $\dim \partial\Gamma = 0$ and Γ is non-elementary if $\#(\Gamma_i/\Gamma_i') > 2$ for $i = 1,2$.

Remark: The above discussion (obviously) extends to amalgamated products of finite families of groups and to HNN-extensions.

3.2.A Proposition: *A word hyperbolic group* Γ *has* $\dim \partial\Gamma = 0$ *if and only if it decomposes into amalgamated products and HNN-extensions of virtually cyclic (i.e. finite extensions of cyclic) groups over finite subgroups.*

This is an immediate corollary of Stalling's theorem on groups with infinitely many ends.

Remark: There is a close link between the space of ends $E\Gamma$ and the hyperbolic boundary $\partial\Gamma$. Namely every metric $|\ |$ on Γ can be transformed in a natural way to another metric, say $|\ |'$ on Γ which makes $(\Gamma, |\ |')$ hyperbolic and such that $(\partial\Gamma, |\ |') = E\Gamma$ in the case where $|\ |$ is a word metric for a finite generating subset in Γ.

3.3 Groups with $\dim \partial\Gamma = 1$: The simplest example is the fundamental group Γ of a closed surface of genus g. This group is word hyperbolic for $g \geq 2$ and $\partial\Gamma$ is homeomorphic to S^1. Conversely, if a word hyperbolic group Γ has $\partial\Gamma \approx S^1$ then Γ is a finite extension of a surface group as a simple argument shows (compare [GM]).

Most small cancellation groups Γ are word hyperbolic and these have dim $\partial\Gamma \leqslant 1$. For example, 1/6-groups are word hyperbolic (see 4.7). But 1/5-groups only are semihyperbolic and instead of the (linear) inequality of 2.3.A they satisfy the *quadratic* isoperimetric inequality $A(f) \leqslant C(L(f))^2$, where the constant $C \geqslant 0$ depends on the presentation.

Let Γ_1 and Γ_2 be word hyperbolic without torsion and let $\Gamma_i' \subset \Gamma_i$, $i = 1,2$ be infinite *maximal* cyclic subgroups (i.e. $\gamma^p \in \Gamma_i \Rightarrow \gamma \in \Gamma_i$ for all $p = 1,2,\ldots,$ and $i = 1,2$). Then (by an easy argument) the amalgamated product $\Gamma = \Gamma_1 \underset{\Gamma'}{*} \Gamma_2$ is word hyperbolic and $\dim \partial\Gamma \leqslant m = \max(1, \dim \partial\Gamma_1, \dim \partial\Gamma_2)$. In fact, $\dim \partial\Gamma = m$, unless Γ_1 and Γ_2 are free products of the form $\Gamma_i = \Gamma_i' * \Gamma_i''$, $i = 1,2$.

Start with an arbitrary finite presentation $\{g_1,\ldots,g_p \mid w_1,\ldots,w_q\}$ (where w_j, $j = 1,\ldots,q$ are some cyclically irreducible words in g_i, $i = 1,\ldots,p$ and the implied relations are $w_j = 1$, $j = 1,\ldots,q$). Then introduce new generators g_j' and g_j'' for $j = 1,\ldots,q$ (which correspond to w_j) and define Γ' by the presentation

$$(*) \qquad \{g_i, g_j', g_j'' \mid w_j = [g_j', g_j'']\}$$

for $i = 1,\ldots,p$ and $j = 1,\ldots,q$, where $[g',g'']$ denotes the commutator $g'g''(g')^{-1}(g'')^{-1}$. It is not hard to see that every such Γ' is word hyperbolic and $\dim \partial\Gamma' \leqslant 1$. The simplest such presentation is $\{g,g',g'' \mid g^k = [g',g'']\}$ for some $k \geqslant 1$ and the corresponding Γ' has $\dim \partial\Gamma' = 1$ for $k \geqslant 2$.

__A__ __geometric__ __explanation__: Let P be a compact connected 2-dimensional cell complex with p 2-cells. Each 2-cell is attached to the 1-skeleton of P by a continuous map of the boundary of the disk, say by $\sigma_j\colon S^1 = \partial D^2 \longrightarrow P^1$, $j = 1,\ldots,p$. Let us replace the j-th 2-cell by a compact connected surface S_j with $\partial S_j = S^1$, attached to P^1 by σ_j, for all $j = 1,\ldots,p$. If the maps σ_j are not contractible (e.g. P is a *simplical* complex) and the surfaces S_j have negative Euler characteristic, then the resulting space, say P', is aspherical and the fundamental group $\Gamma' = \pi_1(P')$ is word hyperbolic with $\dim \partial\Gamma' \leqslant 1$.

(One recaptures the previous Γ' with punctured tori for S_j.) The proof is left to the reader (compare 4.2.D.).

3.4 Hyperbolization of polyhedra: The above geometric construction can be generalized in several ways to n-dimensional polyhedra P for all $n \geqslant 2$. Here is the simplest version of such a "hyperbolization" of an n-dimensional polyhedron P. First we slightly modify the previous construction for $n = 2$. Start with the 2-cube $\square^2 = [-1,+1] \times [-1,+1]$ and let

$$\square_h^2 = (\partial\square^2 \times [-1,+1])/\mathbb{Z}_2,$$

where the involution (i.e. the non-trivial element in $\mathbb{Z}_2$) acts by

$$(t_1,t_2,t_3) \longmapsto (-t_1,-t_2,-t_3).$$

Observe that $\square_h^2$ is homeomorphic to the Möbius band and that the boundary $\partial\square_h^2$ is *canonically* homeomorphic to $\partial\square^2$.

Now let P be a 2-dimensional *cubical* polyhedron. That is P is a union of 2-cubes, where every two cubes meet (if they meet at all) across a common face (of dimension zero or one). Then we remove from P the interior of every cube $\square^2$ and attach instead the "hyperbolized" cube $\square_h^2$ using the canonical homeomorphisms $\partial\square_h^2 = \partial\square^2$. Notice that the fundamental group $\Gamma = \pi_1(P_h)$ of the resulting (hyperbolized) polyhedron P_h can be presented by

$$(**) \qquad \{g_i, g_i' \mid (w_j)^2 = g_j'\}.$$

Not every presentation (**) gives a word hyperbolic group but our groups Γ (related to cubical polyhedra P) are easily seen to be hyperbolic with $\dim \partial\Gamma \leqslant 1$.

Next take the 3-cube $\square^3 = \square^2 \times [-1,+1]$ and observe that the boundary $\partial\square^3$ is a cubical 2-polyhedron (homeomorphic to S^2) and that the involution $(t_1,t_2,t_3) \longmapsto (-t_1,-t_2,-t_3)$ on $\partial\square^3$ induces in an obvious way an involution without fixed points on the "hyperbolized" boundary $(\partial\square^3)_h$ (which is homeomorphic to the connected sum of six projective planes). This involution and $t \longmapsto -t$ on $[-1,+1]$ induce an involution on $(\partial\square^3)_h \times [-1,+1]$ which is used to define

$$\square_h^3 = ((\partial\square^3)_h \times [-1,+1])/\mathbb{Z}_2.$$

Notice that $\square_h^3$ is homeomorphic to the total space of a (non-trivial) bundle with the fiber $[-1,+1]$ over the connected sum of four projective plane and the boundary $\partial\square_h^3$ is *canonically* homeomorphic to $(\partial\square^3)_h$.

Now, let P be an arbitrary cubical polyhedron of dimension 3. First we replace the two skeleton P^2 of P by the hyperbolization P_h^2 and then we attach $\square_h^3$ to the hyperbolized boundary $(\partial\square^3)_h \subset P_h^2$ of every 3-cube $\square^3$ in P using the above canonical homeomorphism. Thus we obtain a certain 3-dimensional polyhedron P_h.

This construction obviously generalizes by induction on $n = \dim P$ to all cubical polyhedra P. Namely each $\square^k$-cube in P for $k = 2,\dots,n$ is replaced as earlier by

$$\square_h^k \overset{\text{def}}{=} ((\partial\square^k)_h \times [-1,+1])/\mathbb{Z}_2.$$

This gives us our hyperbolization $P \to P_h$ for all cubical polyhedra.

It is not hard to see that the polyhedron P_h is aspherical and that it admits a (natural) continuous map $P_h \longrightarrow P$, such that the induced cohomology homomorphism

$$H^*(P;\mathbb{Z}_2) \longrightarrow H^*(P_h;\mathbb{Z}_2)$$

is injective. In particular, if P is a closed manifold then P_h also is a closed manifold which is aspherical and which comes with a map $P_h \longrightarrow P$ of degree $= 1 (\mathrm{mod}\ 2)$.

We shall see in 4.3 that *the fundamental group* $\Gamma = \pi_1(P_h)$ *is word hyperbolic for every cubical polyhedron* P and that

$$\dim \partial\Gamma \leqslant \dim P - 1.$$

In fact,

$$\dim \Gamma = \dim P - 1,$$

provided $H^n(P;\mathbb{Z}_2) \neq 0$ for $n = \dim P$. It will become clear later that $\partial\Gamma$ is homeomorphic to S^{n-1} if P is an n-dimensional *manifold*.

3.4.A An obvious drawback in the above construction is the non-orientability of $\square_h^k$ which is due to the fact that the involution $x \leftrightarrow -x$ on the odd dimensional (hyperbolized) sphere is orientation preserving. Let us describe a (slightly less canonical) orientable hyperbolization. Start with a closed oriented manifold S divided into two submanifolds $S = S_+ \cup S_-$ and let $I: S \longrightarrow S$ be an involution permuting S_+ and S_- and keeping fixed $\partial S_+ = \partial S_- = S_+ \cap S_-$. Denote by $\Delta_h(S)$ the cylinder $S \times [-1,+1]$ with the points $s_+ \times (-1)$ and $s_+ \times (+1)$ identified for all $s_+ \in S_+$. Observe that $\Delta_h = \Delta_h(S)$ in a manifold whose boundary is identical to S.

Now, assume by induction, we have a hyperbolization procedure for all (n-1)-dimensional simplical polyhedra. Take an n-dimensional polyhedron P and let P_h^{n-1} be the hyperbolization of the first baricentric subdivision of the (n-1)-skeleton of P. Then the hyperbolized boundary S_h of every n-simplex Δ^n in P has an involution I as above and we hyperbolize P by attaching $\Delta_h(S_h)$ to S_h for all $\Delta^n \subset P$.

It is easy to see that this hyperbolization P_h is aspherical and that the obvious map $P_h \longrightarrow P$ is injective on $H^*(P)$. But the fundamental group $\pi_1(P_h)$ is not hyperbolic for $n \geqslant 3$, as it contains isomorphic copies of $\mathbb{Z} \oplus \mathbb{Z}$. Yet this group is *semihyperbolic* as P_h admits a (singular) metric of non-positive curvature (see Section 4).

If P is a closed (oriented) manifold then, clearly, P_h also is a closed (oriented) manifold. Moreover, a simple application of (the existence of) local combinatorial formula for Pontryagin classes shows that the monomorphism $H^*(P;\mathbb{Q}) \longrightarrow H^*(P_h;\mathbb{Q})$ sends Pontryagin classes of P to those of P_h. In particular, P_h has the same Pontryagin numbers as P. (This also is clear with 3.4.C.)

3.4.B Now let us construct a hyperbolization which is orientable, preserves Pontryagin classes and has word hyperbolic fundamental group. The inductive step $n-1 \Rightarrow n$ is as follows. First we take a sufficiently fine subdivision P' of P, hyperbolize the (n-1)-skeleton of the subdivision and only then fill in the hyperbolized boundaries of *not* subdivided n-simplices Δ in P. We assume as earlier that the hyperbolization of the (n-1)-skeleton of the subdivided Δ agrees with symmetries of Δ. Then we have an even number of n-simplices of P' in Δ, say Δ_i' and Δ_i'' which (for fine enough subdivision) can be assumed pairwise disjoint and oppositely oriented in Δ. Then we attach to each such pair the cylinder $(\partial\Delta_i')_h \times [-1,1]$ where $(\partial\Delta_i')_h \times (-1) = (\partial\Delta_i')_h$ and $(\partial\Delta_i')_h \times 1$ is identified with $(\partial\Delta_i'')_h$.

3.4.C __Relative hyperbolization:__ Let P_0 be a subpolyhedron in P, take the first baricentric subdivision P' of P and apply a hyperbolization to every simplex Δ^k in P' for $k \geqslant 2$ which is not contained in P_0. Call the resulting polyhedron $H = H(P,P_0)$ and observe that

(1) The inclusion homomorphism $\pi_1(P_0) \longrightarrow \pi_1(H)$ is injective.

(2) If P_0 is aspherical then H also is aspherical. In particular, the fundamental group of every aspherical polyhedron embeds into that of some closed aspherical manifold.

(3) If the group $\pi_1(P_0)$ is semihyperbolic then so is $\pi_1(H)$. Moreover if we use a strict hyperbolization (i.e. the one in 3.4 or 3.4.B) then $\pi_1(H)$ is hyperbolic, provided $\pi_1(P_0)$ is such.

(4) If P_0 is a closed manifold and $P = P_0 \times [0,1]$, then $H(P,P_0 \times 0)$ is a cobordism between P_0 and the hyperbolization P_{0h} of

$P_0 = P_0 \times 1$. In particular the hyperbolizations in 3.4, 3.4.A and 3.4.B preserve Stiefel-Whitney numbers of manifolds, as well as (in cases 3.4.A and B) Pontryagin numbers.

(5) If P_0 is a closed aspherical manifold which is the boundary of a manifold $P \supset P_0$ then H is an aspherical manifold with the same boundary $\partial H = \partial P = P_0$.

3.4.D **Reflection groups:** A group Γ is called a *reflection group* if it admits a presentation of the form

$$\{g_1,\dots,g_p \mid g_i^2 = 1, (g_ig_j)^{d_{ij}} = 1\}$$

for $i,j = 1,\dots,p$, where d_{ij} are given numbers $\in \{0,1,2,\dots\}$ and where the relation $g^0 = 1$ by definition means no relation at all. Every reflection group Γ can be realized by a discrete subgroup in the orthogonal group $O(m,n)$ for some m and n depending on Γ (see [Bour]). It follows that every reflection group is residually finite and virtually torsion free (i.e. it contains a subgroup of finite index without torsion). The geometric and topological significance of reflection groups was pointed out by W. Thurston and then clarified by Davis (see [Dav]) who found, for example, an aspherical manifold V of a given dimension $n \geqslant 4$ whose fundamental group Γ is a subgroup of finite index in a reflection group and such that Γ is not simply connected at infinity.

Davis' groups are semihyperbolic as they act on spaces with $K \leqslant 0$ (see 4.6) and some reflection groups are hyperbolic. It is claimed in Section 4.E of [Gr5] that a hyperbolization similar to the one described in the previous section can be achieved with *reflection* groups. Then L. Siebenmann pointed out to me a difficulty in the construction of hyperbolic reflection groups. Now 4.E of [Gr5] should be regarded as a conjecture. This conjecture is justified by some interesting examples of hyperbolic reflection groups found (but yet unpublished) by Siebenmann and Ancel.

§4. Singular spaces and orbispaces with $K \leqslant 0$

For a geodesic space X the inequality $K(X) \leqslant \chi$ is *defined* (according to Alexandrov, see [Al]) via the CAT-inequality: Every $x_0 \in X$ admits a neighborhood where every three points satisfy CAT(χ) (compare 2.4). If $\chi \leqslant 0$ then the (local!) condition $K \leqslant \chi$ yields the following (well known and easy to prove) global

Theorem (Cartan-Hadamard-Alexandrov): *If* $K(X) \leqslant \chi \leqslant 0$ *and* X *is simply connected, then* X *is contractible and satisfies* CAT(χ) *(for all quadruples of points* x, y, z *and* x_0 *in* X).

Corollary: *Let* f_i: $[0,a_i] \longrightarrow X$ *for* $i = 1,2$ *be locally isometric maps. Then the (distance) function* $|f_1(t_1)-f_2(t_2)|$ *is convex on* $[0,a_1] \times [0,a_2]$. *In particular the maps* f_i *are globally isometric.*

From this one easily derives the following

4.1 Hyperbolicity criteria:

(A) *If* X *is simply connected and* $K(X) \leqslant \chi < 0$ *then* X *is hyperbolic. Moreover* X *satisfies the sharpened* δ-*inequality* (**) *of* 2.4.

(B) *Let* X *be simply connected with* $K(X) \leqslant 0$ *and let the (full) isometry group be cocompact on* X. *Then* X *is hyperbolic unless it receives an isometric map* $\mathbb{R}^2 \longrightarrow X$ (see [B-G-S]).

4.2 Polyhedra with $K \leqslant \chi$: Let X be a finite dimensional simlical complex, such that every k-simplex Δ in X is isometric to a simplex Δ^k in the *model space* (M^k,χ_0) which is the complete simply connected manifold of constant curvature χ_0. A simplex

$\Delta^k \subset M^k$ by definition is a compact subset which is the intersection of (k+1) half-spaces in M^k in general position. Call such an X an (M,x_0)-*simplical* space and observe that the *link* $L_\Delta \subset X$ of every k-simplex $\Delta \subset X$ has a natural (M,1)-structure. Recall that L_Δ is a (k+1)-codimensional (sub)complex whose simplices correspond to (solid angles along Δ of) the simplices $\Delta' \supset \Delta$ in X. The following (well known and easy to prove) criterion reveals the combinatorial meaning of the curvature.

4.2.A *An* (M,x_0)-*simplical space* X *has* $K(X) \leqslant x$ *if and only if* $x \geqslant x_0$ *and* L_Δ *satisfies* CAT(1) *for all* $\Delta \subset X$ *(compare* [Al]*).*

This criterion is especially useful in conjunction with another simple (and well known) fact.

4.2.B *An* (M,x_0)-*simplical space* L *satisfies* CAT(1) *if and only if* $K(L) \leqslant 1$ *and one of the following conditions is satisfied.*

(1) *Every two points* ℓ_1 *and* ℓ_2 *in* L *with* $|\ell_1-\ell_2| < \pi$ *can be joined by at most one segment in* L.

(2) *If the circle* S_ℓ *with the geodesic metric of length* ℓ *admits an isometric map* $S_\ell \longrightarrow L$, *then* $\ell \geqslant 2\pi$.

4.2.C __Remark__: It is often convenient to use a decomposition of X not into simplices but into larger subspaces, called *blocks* $\square \subset X$ which are subpolyhedra for some (M,x_0)-simplical structure on X (e.g. the building cubes of a cubical polyhedron) and which satisfy the following five conditions

(1) Each $\square$ is assigned a subpolyhedron $\partial\square \subset \square$ called the boundary of $\square$ which is the union of some blocks $\square' \subset \square$ with $\dim \square' < \dim \square$.

(2) The complement $\square \setminus \partial\square$ is locally isometric to (M^k, x_0) for $k = \dim \square$.

(3) The intersection of every two blocks is a union of blocks.

(4) If $\square_1 \subset \square_2$ and $\square_1 \neq \square_2$ then $\square_1 \subset \partial\square_2$ for every two blocks $\square_1$ and $\square_2$ in X.

(5) The links L_Δ and $L_{\Delta'}$ of every two k-dimensional simplices Δ and Δ' in $\square$ are isometric for each k-dimensional block $\square$. The isometry class of L_Δ then is denoted by $L_\square$.

Given such a decomposition of X into blocks one can forget the original simplical structure and apply 4.2.A and 4.2.B directly to the (M,k_0)-*block space* X.

4.2.D __Two-dimensional polyhedra__: The condition $K(X) \leqslant \chi$ is easy to verify if $\dim X = 2$. In this case the link of every vertex $x_0 \in X$ is a 1-polyhedron (graph) whose every edge is given length equal the angle of the corresponding triangle in X. Criterion 4.2.B now requires every simple closed curve in the link L_{x_0} to have length $\geqslant 2\pi$.

__Examples__:

(a) Let every simple closed curve in L_{x_0} contain at least $\ell \geqslant 6$ edges. Then the standard simplical metric in X, where all 2-simplices are equilateral Euclidean triangles (with the angles $2\pi/6$), has $K \leqslant 0$. Furthermore, if we replace six by seven and use sufficiently small equilateral hyperbolic triangles with curvature -1 we get a metric on X with $K \leqslant -1$.

(b) Let X be built of regular flat k-gons for $k \geqslant 6$. Then again $K(X) \leqslant 0$ and for $k \geqslant 7$ one gets $K \leqslant -1$.

(c) Let us allow k-gones for $k \geqslant 4$ and add the inequality $\ell \geqslant 4$ for the links of all vertices. Then again we get $K(X) \leqslant 0$. If we additionally assume $\max(k,\ell) \geqslant 5$, then $K(X) \leqslant -1$ for some metric on X.

4.2.C __Cubical polyhedra__: Let X be an n-dimensional polyhedron built of unit Euclidean cubes and let us give a (necessary and sufficient) condition for $K(X) \leqslant 0$. Take the link $L_\square$ of some cube $\square$ in X and say that $L_\square$ satisfies *no-Δ-condition* if for every three vertices in $L_\square$ where every two are joined by an edge in $L_\square$ there exists a 2-simplex in $L_\square$ with these three vertices.

__**Proposition**__: *If* $L_\square$ *satisfies no-Δ-condition for all* $\square$ *in* X *then* $K(X) \leqslant 0$.

__**Proof**__: Criterion 4.2.B and an obvious induction on dimension reduce the problem to the following

__**Lemma**__: *Let* L *be a simplical polyhedron built of regular spherical simplices of curvature* +1 *and of diameter* $\pi/2$. *If* $K(L) \leqslant 1$ *and* L *satisfies no-Δ-condition, then every closed geodesic in* L *has length* $\geqslant 2\pi$.

__**Proof**__: Take the ball $B_v \subset L$ of radius $\pi/2$ around some vertex $v \in L$. This B looks very much like the ordinary hemisphere. Namely, if two points x and y in the boundary ∂B are joined by a geodesic segment s in B of length $< \pi$, then this segment is necessarily contained in ∂B. This is seen by looking at the surface (of curvature $\leqslant 1$) which is the union of the unit segments starting at v and meeting our segment s.

Now, for every closed geodesic γ in L, there exist two simplices Δ_1 and Δ_2 in L such that γ passes from Δ_1 to Δ_2 across a common face in $\Delta_3 \subset \Delta_1 \cap \Delta_2$. Then γ meets the *open* $\pi/2$-balls around some vertices $v_1 \in \Delta_1$ and $v_2 \in \Delta_2$

opposite to Δ_3 and hence, length $\gamma \geqslant 2\pi$.

Remarks:

(a) This argument is similar to the one used in the theory of Tits and Bruhat-Tits buildings which provide most remarkable examples of polyhedra with $K \leqslant 1$ and with $K \leqslant 0$ (see [Ti_1]).

(b) The lemma (and the proof) holds true for non-regular simplices of *size* $\geqslant \pi/2$. This means that the distance from each vertex to the opposite face is $\geqslant \pi/2$. For example, every 3-dimensional polyhedron P built of dodecahedra and satisfying no-Δ-condition has $K \leqslant 0$. In fact, by using hyperbolic dodecahedra with 90-degree dihedral angles one can give P a metric with $K \leqslant -1$.

(c) **Siebenmann's no-$\square$-condition:** This applies to a simplical complex L and claims in addition to no-Δ that every quadrilateral of edges in L bounds a union of two triangles in L.

One easily sees with the proof of the lemma that every closed geodesic of length 2π in L can be deformed to a geodesic consisting of four edges that is a quadrilateral of edges which bounds no pair of triangles. It follows that no-$\square$ ensures a lower bound $\ell \geqslant 2\pi + \epsilon$ for the length ℓ of all closed geodesics in L where $\epsilon > 0$ only depends on dim L.

Corollary: *Let every* $L_\square$ *in* X *satisfies no-Δ-and-no-$\square$-conditions and let us give* X *a* $(M,-\epsilon')$ *geometry, where each cube in* X *is isometric to the unit cube in the hyperbolic space of curvature* $-\epsilon'$. *If* $|\epsilon'| > 0$ *is sufficiently small then* $K(X) \leqslant -\epsilon'$. *In particular, if* X *is compact then the fundamental group* $\pi_1(X)$ *is word hyperbolic.*

(d) Let us indicate without a proof (which is easy and left to the reader) a simple general criterion on deformations of metrics without geodesics that are shorter than 2π.

Proposition: *Let* $|\ |_t$ *be a continuous family of metrics with* $K \leqslant 1$ *on a fixed space* L *such that* $|\ |_{t'} \geqslant |\ |_t$ *for all* t *and* $t' \geqslant t$. *If* L *contains no closed* $|\ |_0$*-geodesics of length* $\leqslant \ell < 2\pi$ *then there is no such* $|\ |_t$*-geodesics in* L *for all* $t \geqslant 0$.

Remark: In many (but not in all) cases the strict monotonicity $|\ |_{t'} > |\ |_t$ ensures "no $|\ |_t$-geodesics of length 2π," provided there is no $|\ |_0$-geodesics of length $\leqslant 2\pi$.

4.3 **Cutting and pasting for** $K \leqslant 0$: Let X have $K \leqslant \chi$ and let $Y \subset X$ be a *locally convex* subspace. That is for every $y \in Y$ there exists an $\epsilon > 0$ such that every (distance minimizing) segment $[x_1,x_2] \subset X$ with $x_i \in Y$ and $|x_i - y| \leqslant \epsilon$ for $i = 1,2$, is contained in Y. Let Γ freely and isometrically act on Y and let $\bar{X} = (X \setminus Y) \cup Y/\Gamma$ with the obvious geodesic metric. Then (by an easy and well known argument) the space $\bar{X}$ also has $K(\bar{X}) \leqslant \chi$.

Example: Let Y consist of two disjoint isometric copies, $Y = Y_1 \cup Y_2$ and $\Gamma \approx \mathbb{Z}_2$ permutes the copies. Then $\bar{X}$ is obtained by gluing Y_1 and Y_2 according to the isometry α: $Y_1 \leftrightarrow Y_2$ for the generator $\alpha \in \mathbb{Z}_2$. Algebraically speaking, this gives us amalgamated and HNN products in the category $K \leqslant 0$.

4.3.A The above criterion immediately shows that the natural (M,0)-metrics in the hyperbolized polyhedra P_h in 3.4, 3.4.A and 3.4.B have $K \leqslant 0$ as they are built of standard blocks attached along their locally convex boundaries. Furthermore one can easily perturb the metrics on P_h of 3.4 and 3.4.B to a one with $K < 0$ which implies the hyperbolicity of $\pi_1(P_h)$ when P is compact.

4.4 **Ramified coverings**: Take again a locally convex $Y \subset X$ and a covering $\tilde{X}^*$ of the complement $X \setminus Y$. If $\tilde{X}^*$ is a trivial covering, that is a disjoint union of several copies of $X \setminus Y$, then we can glue together the corresponding copies of X along Y as in 4.3 and obtain a space $\hat{X}$ with $K(\hat{X}) \leqslant \chi$, provided $K(X) \leqslant \chi$. Now we generalize this construction to non-trivial coverings $\tilde{X}^*$ by letting $\hat{X} = \tilde{X}^* \cup Y$. This $\hat{X}$ carries a natural geodesic metric induced by the map $p: \hat{X} \longrightarrow X$ which is the covering map on $\tilde{X}^*$ and the inclusion on Y. A simple application of 4.3 shows this metric has $K \leqslant \chi$.

__Example__: Let X be a manifold and $Y \subset X$ a codimension two submanifold. Then $\hat{X}$ can be obtained as the metric completion of $\tilde{X}^*$ with the metric induced from X. The above criterion applies if X is Riemannian manifold and Y is a totally geodesic (i.e. locally convex) submanifold. Such submanifolds $Y \subset X$ are abundant if X is compact of constant negative curvature with an *arithmetic* fundamental group. (This leads to useful examples of ramified coverings $\hat{X}$ in [G-T].) Notice that completions $\hat{X}$ of *infinite* coverings $\tilde{X}^*$ of $X^* = X \setminus Y$ provide interesting examples of not locally compact spaces with $K \leqslant 0$.

4.4.A __**Multiple ramifications**__: There are many cases where $Y \subset X$ is *not* locally convex but yet some ramified coverings $\hat{X}$ have $K \leqslant 0$, provided $K(X) \leqslant 0$. The standard example is the map $\mathbb{C}^n \longrightarrow \mathbb{C}^n$ given by $(z_1,\ldots,z_n) \longmapsto (z_1^{i_1},\ldots,z_n^{i_n})$ for some integers $i_1,\ldots,i_n$. The (singular) metric on the source induced from the flat metric on the target has $K \leqslant 0$.

Now take an *immersed* codimension two submanifold $Y \subset X$ with transversal selfintersections (normal crossings). A ramified covering $\hat{X}$ is called *normal* if it is locally modelled by the above example for dim X *even*. If dim X is odd we use for the model the Cartesian product of the above map $\mathbb{C}^n \longrightarrow \mathbb{C}^n$ by the identity $\mathbb{R} \longrightarrow \mathbb{R}$.

Now let X be given a Riemannian metric with $K \leqslant \chi$ and let at every selfintersection point of Y different branches of Y are mutually orthogonal (like transversal coordinate subspaces of codimension two in $\mathbb{R}^n$). Then (by an easy argument) the induced metric in every normal ramified covering $\hat{X}$ also has $K(\hat{X}) \leqslant \chi$.

__**Example**__ **(Compare [Gr1])**: Let X be a flat torus and Y a union of flat codimension two subtori. If these subtori meet orthogonally we have $K \leqslant 0$ for normal ramified coverings $\hat{X}$. Furthermore, if these subtori generate $H_{n-2}(X)$, then, by 4.1.B the fundamental group $\pi_1(\hat{X})$ is hyperbolic, provided $\hat{X}$ *nontrivially* ramifies over each subtorus. If dim X is odd, then every generic homomorphism $X \longrightarrow S^1$ induces a *fibration* $\hat{X} \longrightarrow S^1$ whose fiber F is an even dimensional torus normally ramified over flat subtori. Yet these subtori do not meet orthogonally (whenever they meet). In fact, if dim $F \geqslant 4$, then the fundamental group $\pi_1(F)$ is *not* hyperbolic by 5.4.

4.4.B <u>Ramifications associated with finite reflection groups</u>: Let G be a finite group of orthogonal transformations of $\mathbb{R}^n$ such that the quotient $\mathbb{R}^n/G$ is homeomorphic to $\mathbb{R}^n$. Then the map $\mathbb{R}^n \longrightarrow \mathbb{R}^n/G = \mathbb{R}^n$ can be viewed as a ramified covering whose singular locus is the image of fixed points in $\mathbb{R}^n$ of non-identity elements $g \in G$. The standard examples come from groups G' generated by reflections on S^{n-1} and having a simplex for a fundamental domain. Indeed, the subgroup $G \subset G'$ of the orientation preserving transformations has $\mathbb{R}^n/G \approx \mathbb{R}^n$.

Next, one takes a subset Y in an n-dimensional manifold X which locally looks like the ramification locus in $\mathbb{R}^n \approx \mathbb{R}^n/G$ for a given G and then one has a (possibly empty) class of ramified coverings which ramify like the model map $\mathbb{R}^n \longrightarrow \mathbb{R}^n/G$ (an appropriate formalism for such a description is provided by the language of orbifolds, see 4.5).

<u>Example</u>: Let $G \approx \mathbb{Z}_2 \oplus \mathbb{Z}_2$ act on $\mathbb{R}^3$ by $(x,y,z) \longmapsto (-x,-y,z)$ and $(x,y,z) \longmapsto (x,-y,-z)$. Then the ramification locus in $\mathbb{R}^3 = \mathbb{R}^3/G$ consists of three rays from the origin. Now, we take a graph Y in a 3-manifold X, such that there are exactly tree edges at every vertex of this graph. The linear ramification pattern for the above $\mathbb{R}^3 \longrightarrow \mathbb{R}^3/G$ gives a nice class of ramified covers $\hat{X} \longrightarrow X$. If, for example, the manifold X has $K \leqslant 0$, if the graph Y consists of *geodesics* between the vertices in Y and if these geodesics meet at 120-degree angles at the vertices, then (by an easy argument) $\hat{X}$ also has $K \leqslant 0$. We suggest to the reader to find specific examples of such Y in the flat torus T^3 and then look at the standard series (A_n, B_n etc.) of reflection groups.

4.5 **Orbifolds and orbispaces**: Let a group Γ act discretely and isometrically on a (Riemannian) manifold X. If the action is free then Γ can be recaptured from X/Γ as $\Gamma = \pi_1(X/\Gamma)$. The same can be done for non-free action with an additional *orbistructure* on X/Γ which takes into account the (finite) isotropy subgroups $\Gamma_x \subset \Gamma$ for all $x \in X$. Namely, if $x \longmapsto v \in V = X/\Gamma$, then a small neighborhood $U_v \subset V$ of v is naturally represented as $\tilde{U}/\Gamma_x$ for some neighborhood $\tilde{U} \subset X$ of x.

Now we *start* with a space V, a collection of groups Γ_v for all $v \in V$ operating on some connected manifolds X_v, and maps of X_v/Γ_v onto some neighborhoods $U_v \subset V$, such that these data agree in the way suggested by the example of $V = X/\Gamma$ (see [Th] for details).

We are interested in a more general case where X_v need not be manifolds but rather general geodesic spaces. The *orbispace* V (called *orbifold* if X_v are manifolds and *orbihedron* if X_v are polyhedra) also comes with a geodesic metric and the maps $X_v/\Gamma_v \longrightarrow U_v$ are assumed isometric. The curvature inequality $K(V) \leqslant \chi$ by definition is: $K(X_v) \leqslant \chi$ for all spaces X_v.

In the orbifold category the orbistructure is usually determined by the metric. For example, the unit segment [0,1] carries a unique (maximal) orbifold structure as $[0,1] = \mathbb{R}/\Gamma$ for Γ generated by integral translations and $t \longleftrightarrow -t$. Yet [0,1] has many orbihedral structures. Namely if we divide any infinite regular tree (by its full isometry group (which is not discrete though totally disconnected)), we also get [0,1] but with another orbistructure.

A *universal covering* of an orbispace V is a simply connected space X with an action of Γ such that X/Γ equals Y in the orbispace sense. The group Γ is called *a fundamental group* of V. Notice that neither existence nor uniqueness of Γ is automatic. Yet in the orbifold case X and Γ are unique, though they do not always exist.

If V is an orbivolds with $K(V) \leqslant 0$ then the universal covering

X (which is a manifold with $K \leqslant 0$ in the usual sense) does exist by a trivial argument, but this is unknown for orbihedra. However, there is an additional *rigidity* condition (ruling out tree-like behavior) which ensures the existence and uniqueness of the universal covering. Namely, V is called a *rigid orbispace* if none of the spaces X_v admits a non-trivial isometry which fixes a non-empty open subset in X_v. In the rigid case the uniqueness of X and Γ is obvious and the inequality $K(V) \leqslant 0$ ensures the existence.

Remark: Non-rigid orbihedra, especially those with $K \leqslant 0$, (for example trees) are very interesting objects with abundance of symmetries. Some examples of them can be found in 4.C″ and C‴ of [Gr5].

Unrestricted orbihedra: A geodesic metric space V is called *unrestricted* if every local geodesic can be extended. That is every locally isometric map $[a,b] \longrightarrow V$ extends to a locally isometric map $\mathbb{R} \longrightarrow V$ for all $[a,b] \subset \mathbb{R}$. Then one defines *unrestricted orbispaces* in terms of local extension of geodesics in the corresponding spaces X_v.

If V is a polyhedron with $K \leqslant 0$, then "unrestricted" is (obviously) equivalent to the following property of the links L_Δ of V: for each point $\ell \in L_\Delta$ there exists an *opposite* point ℓ' in L_Δ, such that $|\ell-\ell'|_{L_\Delta} \geqslant \pi$. (If ℓ and ℓ' lie in different components of L_Δ we assume $|\ell-\ell'|_{L_\Delta} = \infty$.)

Every compact unrestricted (orbi)space V with $K \leqslant 0$ obviously has *infinite* universal covering and hence $\pi_1(V)$ also is infinite. Furthermore, if $K < 0$ then $\pi_1(V)$ is a *non-elementary* hyperbolic group unless V is isometric to S^1 or to $[0,1]$ with the standard orbistructure.

4.5.A Examples of orbihedra:

(a) Start with a two dimensional polyhedron V and give it the orbistructure which is *supported* in the centers of the triangles in V (the support of an orbistructure is the set of those $v \in V$ where

Γ_v is non-trivial) and equals there to the $2\pi/k$-*structure*. That is the disk in $\mathbb{R}^2$ divided by the usual $\mathbb{Z}_k$-action. If $k \geq 2$ for all triangles, then the obvious (orbi)metric on V has $K \leq 0$ and if $k \geq 3$ then $K < -\epsilon^2 < 0$. The fundamental groups Γ of these V provide plenty of hyperbolic groups.

(b) Let V be a manifold and V' an immersed submanifold of codimension two with transversal self-intersections. At every p-multiple intersection point we give V the orbistructure of the standard action of $\mathbb{Z}_{i_1} \times \mathbb{Z}_{i_2} \times \ldots \mathbb{Z}_{i_p}$ on $\mathbb{C}^p \times \mathbb{R}^m$, where $2p + m = \dim V$ (compare 4.4.A). In order to define an orbistructure we assume an obvious agreement between p and (p+1)-multiple points. For example if V' is the union of p embedded submanifolds then such a structure is given by assigning a ramification index i_j for $i = 1,\ldots,p$ to each submanifold. Now if V has a metric with $K \leq 0$ and V' is geodesic with *orthogonal* crossings then same metric viewed as orbimetric also has $K \leq 0$ and hence the *orbifold* V admits the universal covering, that is the maximal *ramified* covering of the *manifold* V with given ramification along V'.

4.5.B __Cutting and pasting orbispaces:__ This can be done for $K \leq 0$ the same way as for ordinary spaces since the condition $K \leq 0$ is local.

__Example:__ Let V be an orbispace with $K \leq 0$ and let $S \subset V$ be a closed geodesic of length $2\pi r$ which does not meet the support of the orbistructure. (S is the image of locally isometric map of $2\pi r$-circle into V.) Next take the disk $D^2 \subset \mathbb{R}^2$ of radius kr for some $k = 1,2,\ldots,$ divided by the standard action of $\mathbb{Z}_k$ and attach $D^2/\mathbb{Z}_k$ to V by an isometry between $\partial D^2/\mathbb{Z}_k$ and S. The resulting orbifold $V \cup (D^2/\mathbb{Z}_k)$ fails to have $K \leq 0$ since $\partial D^2/\mathbb{Z}_k$ is not locally convex (geodesic) in $D^2/\mathbb{Z}_k$. Yet it becomes such in the limit for $k \longrightarrow \infty$ as $D^2/\mathbb{Z}_k$ converges to the product of $\mathbb{R}_+$ by the $2\pi r$-circle S and the space $V \cup (\mathbb{R}_+ \times S)$ does have $K \leq 0$. This follows from 4.3 if S has no self-intersection points in V and an obvious modification of 4.3 allows such points.

Now, let $K(V) < 0$ and show for large $k > k_0 = k_0(V,S)$ that there is a small perturbation of the metric on $V \cup (D^2/\mathbb{Z}_k)$ which achieves $K < 0$. Assume for simplicity's sake that V is an orbihedron with a polyhedral $(M,-\epsilon^2)$-type metric. We may assume (subdividing the polyhedron if necessary) that S contains a vertex v of the triangulation and two adjacent edges. Denote by v' and v" the corresponding vertices in the link L_v and observe that the (angular) distance between v' and v" satisfies $|v'-v"| \geqslant \pi$. Since $K(V) < 0$ there is a small perturbation of the metric on V, such that the new distance in the link satisfies $|v'-v"|_{new} \geqslant \pi+\delta$ for a small $\delta > 0$. Now, instead of D^2 we take the regular k-gone D_k with edge length $2\pi r$ in the hyperbolic plane with (constant) curvature $-\chi < 0$. If k is large and $|\chi|$ is small then the angles α in D_k between adjacent edges satisfy $|\pi-\alpha| \leqslant \delta$. Therefore, if we attach $D_k/\mathbb{Z}_k$ to V along S, such that the corner point in $\partial(D_k/\mathbb{Z}_k)$ goes to v, then the resulting space will have $K < 0$.

__Remark:__ The above perturbation easily generalizes to more general orbispaces where a small amount of positive curvature can be engulfed into surrounding negative curvature. For example, one could take a higher dimensional locally convex (immersed) subspace V' in V rather than S and attach orbicones to appropriate coverings of V'.

4.5.C __Construction of infinite torsion groups__: Start with a compact unrestricted polyhedron V with $K < 0$, for example with the figure ∞ or with a surface of constant negative curvature. Take a closed geodesic in V, attach the $2\pi/k_1$-orbicone as above and obtain an orbihedron V_1 with $K(V_1) < 0$ which is obviously unrestricted. Do the same to V_1 and get V_2. Notice that some geodesic in V_1 may pass through the point supporting the orbistructure. To avoid (not at all serious) trouble we perturb our metrics in order to remove geodesics from orbisupports. Now, we continue the process by keeping attaching orbicones and obtain a sequence of unrestricted orbispaces with $K < 0$, say $V \subset V_1 \subset V_2 \subset \ldots V_i$, such that (if we wish) the length of the shortest closed geodesic in V_i, say ℓ_i, satisfies $\ell_i \longrightarrow \infty$ for $i \longrightarrow \infty$. It follows that the fundamental group Γ_∞ of the orbispace $V_\infty = \bigcup_i V_i$ is pure torsion. Furthermore, since V_∞ has $K(V_\infty) \leqslant 0$ and is rigid, the universal covering $\tilde{V}_\infty \longrightarrow \tilde{V}$ does exist and has infinitely many ramifications of orbitrarily large order. Thus the group Γ_∞ is infinite. Finally, Γ_∞ is the factor group of $\pi_1(V)$ and hence it is finitely generated. Q.E.D.

__Remark:__ If we start this construction with a lattice in Sp(2,1) we obtain an infinite torsion group with Kazdan's T-property.

__Question:__ Let a group Γ be obtained from a non-elementary word hyperbolic group Γ_0 by adding "on random" infinitely many relations to Γ_0. Does Γ satisfy T-property?

4.6 __Reflection orbihedra with__ $K \leqslant 0$: To grasp the idea start with the unit cube V obtained by dividing $\mathbb{R}^n$ by the group Γ generated by integral translations of $\mathbb{R}^n$ and reflections in the coordinate hyperplanes. The space $\mathbb{R}^n$ can be reconstructed as the universal (orbi)covering of V which is obtained in this case by repeated reflections of V around codimension one faces. The fundamental group Γ ov V is a reflection group (see 3.5) whose generators γ_i (with $\gamma_i^2 = 1$) $i = 1,\ldots,2n$, correspond to the codimension one faces of V. The relations are $(\gamma_i\gamma_j)^2 = 1$ for those pairs of faces which meet at codimension two faces of V.

Notice that the orbistructure of V is supported on the boundary and it is determined by the combinatorial pattern of the codimension one faces. If we take the *nerve* L of the covering of ∂V by these faces, then each face can be identified with the star of a vertex of L in the first barycentric subdivision L' of L while V becomes the cone over L'.

Now we *start* with an *arbitrary* simplical polyhedron L, cover L by the stars of the vertices of L in the first baricentric subdivision L' of L and consider the cone M over L' with the orbifold structure induced by the cover of L by the stars. Notice that the cone over a baricentrically subdivided simplex can be naturally identified with the standard cube. Thus M has a natural structure of a cubical complex and the discussion in 4.2.C applies to the natural (M,0)-cubical metric on M. This metric has $K \leqslant 0$ (in the orbifold sense) if L and the link of each simplex in L satisfy no-Δ-condition. Moreover, if no-$\square$ is satisfied as well, then $K < 0$ for a small hyperbolic perturbation of the (M,0)-metric.

__Corollary:__ *The reflection group* $\Gamma = \Gamma(L)$ *generated by* γ_v *with* $\gamma_v^2 = 1$ *for all vertices* $v \in L$ *and with relations* $(\gamma_v\gamma_w)^2 = 1$ *for all edges* $[v,w]$ *in* L *operates on some polyhedron* X *(that is the universal covering of the orbifold* M) *with* $K \leqslant 0$, *provided no-*Δ *is satisfied. Moreover no-*$\square$ *implies* $K < 0$ *and* Γ *is hyperbolic if* L *is compact.*

__Remark:__ The baricentric subdivision of *every* simplical complex satisfies the no-Δ-condition but the no-$\square$ is harder to get as was pointed out to me by L. Siebenmann.

__Examples:__ Let L be a triangulation of the sphere S^{n-1}. Then, obviously, X is a manifold. More generally, let L be a triangulation of a boundary ∂W of some n-dimensional manifold. The partition into stars gives an orbifold structure to W supported on ∂W. This structure usually has no metric with $K \leqslant 0$ but the corresponding structure on the cone M over ∂W does carry such a structure in the no-Δ-case. It follows that the orbifold W admits a universal covering with the fundamental group $\sqcap = \sqcap(W)$ which comes with a natural epimorphism onto the fundamental group $\Gamma = \sqcap(M)$. Of course, if W is simply connected as a topological space, then $\sqcap(W) = \sqcap(M)$. In general, the (ordinary) fundamental group $\pi_1(W)$ embeds into $\sqcap(W)$ and it is annihilated by the homomorphisms $\sqcap(W) \longrightarrow \sqcap(M)$.

4.7 __Small cancellation polyhedra of dimension two__: There are various combinatorial conditions for 2-dimensional orbihedra (in particular, polyhedra) V which are more general than $K < 0$ (or

$K \leqslant 0$) but yet sufficient for hyperbolicity (or semihyperbolicity). First notice that $K < 0$ (and $K \leqslant 0$) is expressed in terms of the geometry of the links L_v which only depends on homothety classes of the triangles in V. Now, we define a *conformal structure* in a simplical 2-polyhedron V by assigning a real number, called an angle, to every corner of each triangle Δ in V, such that the sum of the three angles of every triangle equals π. These angles define *lengths* (positive or negative) of curves in the links L_v and $K < 0$ ($K \leqslant 0$) means that every closed curve in L_v has length $> 2\pi$ ($\geqslant 2\pi$) for all $v \in V$. Notice, that the existence of a conformal structure with $K < 0$ (or $K = 0$) amounts to solvability of a system of *linear* inequalities in unknown angles.

Similarly one defines 2-orbihedra with conformal $K < 0$ (or $K \leqslant 0$) and easily shows that $K < 0$ ensures hyperbolicity of the fundamental group, while $K \leqslant 0$ provides (some kind of) semihyperbolicity.

4.7.A Now, let us formulate the usual small cancellation conditions for 2-orbihedra. Here, our V is a cell complex, such that

(1) The 1-skeleton V^1 of V has no redundancy: there are at least three branches at every vertex in V^1.

(2) The attaching maps for the 2-cells D in V are topological immersions (i.e. locally injective) $\partial D \longrightarrow V^1$.

(3) The orbistructure (if there is any) is supported at finitely many points in the interiors of 2-cells D, where each D contains at most one $2\pi/k$-point for some $k = k(D)$.

A *one layer plane diagram belonging to* V is given by the following data.

1. A closed topological disk $B \subset \mathbb{R}^2$ subdivided into (n+1) convex k_i-gones for $k_i \geqslant 3$, called $D_0, D_1, \ldots, D_n$, such that

(a) Every two k_i-gones meet (if at all) at a common *vertex* or *edge*, and each D_i for $i \geqslant 1$ does meet D_0.

(b) The *central* k_0-gone D_0 does not meet the boundary ∂B of B and each D_i for $i \geqslant 1$ does meet ∂B.

2. A continuous map $f: B \longrightarrow V$ such that

(i) The map f sends the vertices of the k_i-gones to some vertices in V and the edges go to the 1-skeleton V^1 of V. Furthermore, f is an immersion (i.e. locally one-to-one) on some neighborhood $U \subset B$ of the union of the edges (that is $\bigcup_i \partial D_i$).

(ii) The interior Int D_i is mapped onto the interior Int D of some 2-cell D in V. This map between open disks is a ramified covering of order $k = k(D)$. Thus, f is a cyclic k-sheeted covering of Int D_i minus a point onto Int D minus the orbipoint for all $i = 0,\dots,n$.

Assign to each vertex v_j of D_0, $j = 1,\dots,k_0$, the angle $\alpha_j = 2\pi/\ell_j$ where ℓ_j denotes the number of k_i-gones adjacent to v_j (notice that $\ell_j \geqslant 3$) and define the curvature (excess) of B by

$$K(B) = 2\pi + \sum_{j=1}^{k_0} (\alpha_j - \pi).$$

Notice that $K(B) \leqslant 0$ for $k_0 > 5$ (the familiar 1/5-condition) and $K(B) < 0$ for $k_0 > 6$.

Now, we summarize various small cancellation conditions (for the orbihedra corresponding to a presentation of a group Γ) in the following

Definition: The combinatorial curvature $K_c(V)$ is < 0 if $K(B) < 0$ for all diagrams belonging to V and $K_c(V) \leqslant 0$ if $K(B) \leqslant 0$.

The basic theorem of the small cancellation theory claims that $K_c < 0$ implies *the hyperbolicity of* $\Gamma = \pi_1(V)$.

Sketch of the proof: Let S be an immersed circle in V^1 and let B'